

Week 2 Material - Vector-Valued Functions

7. ¹ The position of a particle is defined by the vector-value function $\vec{r}(t) = \langle \sqrt{t-1}, \frac{t}{2} \rangle$
- Eliminate the parameter to find a Cartesian equation of the curve. Express your final answer in the form $y = f(x)$.
 - Sketch the graph of the Cartesian equation you found in (a).

Once you have drawn the graph:

- Clearly indicate what part of this graph the function $\mathbf{r}(t)$ actually passes over, and
- draw arrows to indicate the direction of motion.

8. ² If two objects travel through space along two different curves, it is often important to know whether they will collide. (Will a missile hit its moving target? Will two aircraft collide?) The curves might intersect, but we need to know whether the objects are in the same position *at the same time*. Suppose the trajectories of two particles are given by the vector functions:

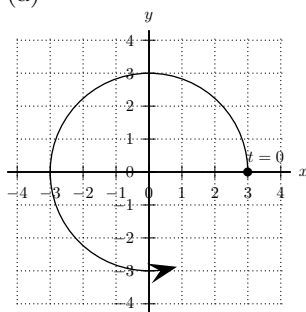
$$\mathbf{r}_1(t) = \langle t-1, 3t-t^2 \rangle, \quad \mathbf{r}_2(t) = \langle -t-1, \frac{1}{4}t^2 + 2t \rangle.$$

- Do the particles collide, and if so when? Provide the collision location as well.
 - Would the paths of the particles intersect, and if so what are those intersection point(s), and when would each of them pass through those intersection point(s)?
9. Consider the curve defined by the equation

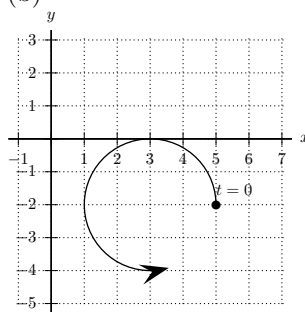
$$(x-3)^2 + (y-7)^2 = 36$$

- Sketch this curve.
 - Find a vector-value function $\mathbf{u}(t)$ that would follow that curve over time.
 - Show by a calculation that your $\mathbf{u}(t)$ always lies on the given curve.
10. ³ For each of the circles shown below, write down a parametric function that would trace out that trajectory, and match any additional properties given. Multiple answers are possible.

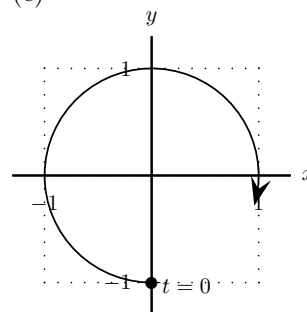
(a)



(b)



(c)



¹Source: previous APSC 171 Test 1

²Similar to previous APSC 171 Test 1

³Source: previous APSC 171 Test 1